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Resource subsidies from adfluvial fishes increase stream productivity

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SUMMARY

1. Anadromous fishes are well known to shape the structure and function of recipient ecosystems by introducing nutrients and rich organic matter from the ocean. In contrast, the importance of potamodromous migrations, confined to freshwater, and the subsidies they provide to stream ecosystems has received much less attention.
2. Our objective was to determine the importance of excretion, eggs, milt, and carcasses as nutrient and energy sources from a large population (82 449 suckers) of migrating longnose (*Catostomus catostomus*) and common white (*Catostomus commersonii*) suckers into a small (wetted width *c.* 10 m) oligotrophic river system. We hypothesise that the adfluvial suckers provide a large material subsidy that increased the productivity of the Cypress River and that this resource subsidy rivals or exceeds those delivered by other native and non-native fishes (e.g. Pacific salmonids).
3. In total there was an estimated 5635 kg of eggs, 2025 kg of milt, and 1 kg of carcasses from suckers that spawn in the Cypress River. Relative to other mainly non-native fishes, suckers provided 92% of the annual egg biomass and 95% of the milt. Suckers, however, only provided <1% (1 kg) of the annual carcass biomass, whereas, pink salmon provided 50% (600 kg). Overall, suckers provided 84% and 78% of the annual subsidies of N and P, or 212 and 14 kg respectively. Ambient NH_4 concentrations in the river were consistently below that predicted from excretion equations suggesting that microorganisms may have rapidly taken up much of the released ammonium.
4. Downstream of the falls, epilithon biomass was over nine times more abundant, benthic invertebrate densities were approximately two times higher, and fish biomass was eight times greater compared to upstream. There were no upstream–downstream differences in substrate organic matter biomass. Fishes downstream of the falls had higher $\delta^{13}C$ and $\delta^{15}N$ values than biota upstream of the falls consistent with the anticipated effect of lake derived subsidies. Using stable isotopes values and a mixing model, we estimated that sucker eggs comprised 25–58% of the diet of stream fishes during the growing season.
5. Suckers provided a large subsidy, greater than all other fishes, without mass mortality and significantly enhanced the productivity of the recipient river system. Stable isotopes revealed that subsidies were incorporated into the stream food web. Differences between salmon and sucker subsidies related to the type, magnitude, timing, and the life history of the donor highlight the potential importance of spawning migrations of adfluvial suckers, and other potamodromous fishes, in streams and rivers around the world.

Keywords: excretion, fragmentation, migratory fish, stable isotopes, subsidy, trophic

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Introduction

The transfer of nutrients and materials across ecological boundaries can strongly influence the productivity and community dynamics of the recipient ecosystem (Polis, Power & Huxel, 2004; Baxter, Fausch & Saunders, 2005). Resource subsidies are a common phenomenon across many ecosystem types involving different taxa including stream invertebrates (Nakano & Murakami, 2001), Pacific salmon (Bilby, Fransen & Bisson, 1996), geese (Giroux *et al.*, 2012), sea turtles (Bouchard & Bjorndal, 2000) and cisco (*Coregonus artedi*) in the Great Lakes (Stockwell *et al.*, 2014) with more examples discovered each year.

Inter-ecosystem transfers of nutrients and material that are important to stream ecosystems and their inhabitants originate from several key sources, including well-studied terrestrial landscape and marine linkages, and lesser understood linkages from fishless stream and lake ecosystems. Terrestrial habitats have long-been recognised for providing nutrients, dissolved and particulate organic carbon that support basal and invertebrate production in streams (Minshall, 1967; Kaushik & Hynes, 1971; Vannote *et al.*, 1980). More directly, terrestrial invertebrates may represent a high quality, but seasonal, food source for fishes which can greatly enhance productivity (Allan *et al.*, 2003; Baxter *et al.*, 2005). Fishless tributary streams represent sources of nutrients, organic matter, and invertebrate prey for fishes in downstream reaches (Wipfli & Gregovich, 2002; Wipfli & Baxter, 2010). The epilimnetic water of lakes provides a rich mix of phytoplankton, zooplankton, bacterioplankton and dissolved organic matter that subsidises downstream reaches which often support an abundance of filtering benthic invertebrates (Richardson & Mackay, 1991; McCreadie & Robertson, 1998; Jones, 2010).

Spawning salmon shape the structure and function of recipient ecosystems and provide nutrients and energy rich organic matter from the ocean environment (Cederholm *et al.*, 1999; Gresh, Lichatowich & Schoonmaker, 2000; Gende *et al.*, 2002). Nutrients from salmon are incorporated into all trophic levels in streams, including riparian vegetation, algae, invertebrates and fish, directly through the consumption of salmon eggs and carcasses and indirectly through excretory products and the decomposition of carcasses (Willson, Gende & Marston, 1998; Gende *et al.*, 2002; Hicks *et al.*, 2005; Helfield & Naiman, 2006). The flow of energy from salmon subsidies can greatly enhance stream productivity resulting in increased density, growth rate, condition, overwintering survival, and ultimately, better marine survival of juvenile salmon than would be expected without migratory salmon subsidies (Wipfli *et al.*, 2003; Bentley *et al.*, 2012; Armstrong & Bond, 2013).

In contrast to marine subsidies associated with migrations of salmon, the importance of potamodromous migrations, confined to freshwater, and the subsidies they provide to stream systems has received much less attention. In freshwater ecosystems, lakes are functionally analogous to oceans for adfluvial fishes [e.g. salmon and trout (Salmonidae), suckers (Castomidae) and smelts (Osmeridae)]. The significance of lake-derived subsidies has focused on excretion and nutrient levels in response to the aggregation of spawning fishes in streams (Ivan, Rutherford & Johengen, 2011; Childress & MacIntyre, 2014; Wheeler, Miller & Crowl, 2014). For example, Childress & MacIntyre (2014) noted that excretion from iteroparous suckers supplies a substantial portion of ecosystem nutrient demand without the need for large numbers of decaying carcasses.

There are few examples of how lake-derived materials (e.g. eggs and carcasses) have influenced community dynamics and productivity in connected stream systems. Eggs represent small, often immediately consumable, sources of nutrients and energy for fishes; whereas, carcasses decay to release nutrients, although Pacific salmonids will directly consume flesh from carcasses. In Alaska, eggs may provide resident stream fishes much of their annual energy surplus (Moore, Schindler & Ruff, 2008; Armstrong & Bond, 2013). Focusing on the eggs of suckers, Childress & MacIntyre (2014) suggested that significant mortality is not required for potamodromous species to act as significant material subsidies, challenging the view of Flecker *et al.* (2010). Further, Childress, Allan & McIntyre (2014) noted that migrating suckers elevated stream nutrients resulting in increased algal accrual and a 12% increase in caddisfly growth.

Resource subsidies stand to have the greatest impact when available nutrients and energy are low in the receiving ecosystem, the magnitude of the pulse is large, the subsidy materials are effectively retained in the recipient ecosystem, and consumers can easily capitalise on the resources (Flecker *et al.*, 2010; Bentley *et al.*, 2012). Oligotrophic streams and rivers exist throughout the world and stand to benefit greatly from subsidies. In support of this, Rand *et al.* (1992) suggested that light appears to limit stream primary productivity more than nutrients from decaying salmon in two already productive rivers. In contrast, Childress & MacIntyre (2014) found that excretion and sucker eggs (immediately available for consumption), represented significant nutrient

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inputs during spawning, even in catchments with elevated nutrient loads from agriculture. Collectively this may suggest that egg subsidies can increase the productivity of streams regardless of their nutrient status (i.e. high or low), whereas, nutrient subsidies from carcasses only increase productivity in nutrient limited oligotrophic streams.

Suckers are one of the most widespread and diverse groups of fishes in North America occupying many aquatic habitats, particularly rivers (Scott & Crossman, 1973; Cooke *et al*., 2005; Quist & Spiegel, 2012). Migrations of suckers can be very large and even rival or exceed runs of Pacific salmon species (Raney & Webster, 1942; Linderman *et al*., 2004; Cooke *et al*., 2005). In contrast to trout and salmon, many suckers do not build a redd or guard eggs after mating, and thus, eggs are more available to consumers (Scott & Crossman, 1973). Although spawning-related mortality is low and does not result in many carcasses, their excretory products and eggs stand to provide a subsidy perhaps surpassing that of migrating trout and salmon in freshwater systems.

Our objective in this study was to determine the importance of excretion, eggs, milt, and carcasses as nutrient sources from a migration of suckers into an oligotrophic river. We hypothesise that adfluvial longnose (*Catostomus catostomus*) and white (*Catostomus commersonii*) suckers provide a material subsidy and increase the productivity of the Cypress River, a tributary to Lake Superior. We also hypothesise that this resource subsidy rivals or exceeds those delivered by other native and non-native fishes (e.g. Pacific salmonids). Testing these hypotheses required an extensive set of methods. We quantified the numbers of suckers ascending the Cypress River to determine the amount of eggs, milt, and nutrients added to the river. We determined the drift rates of eggs available to stream fishes for consumption and the amount of soluble reactive phosphorus (SRP) and ammonium added through excretory products and milt from male suckers. We then assessed the standing biomass of epilithon, course particulate organic matter (CPOM), benthic invertebrates, and fishes. Tissue samples were collected for stable isotope analysis to understand if the river food web exhibited a isotopic shift consistent with our hypothesised subsidy. Lastly, we estimated the subsidy contribution of other fishes and compared these to subsidies provided by suckers, with the understanding that stream productivity enhancement and stable isotope ratios are the integration of all subsidies.

Methods

Bordering Canada and the United States, Lake Superior is the largest of the Laurentian Great Lakes in surface area and depth. It is highly oligotrophic, cold and dominated by native species. The ecology of the Cypress River (N 48.928°, W 87.868°) is representative of other rivers along the north shore of Lake Superior with Precambrian bedrock-dominated basins (Bond & Jones, 2013). With a catchment area of 165 km^2, it experiences a predictable nival and perennial flow regime and its waters are oligotrophic, slightly acidic, and clear with some humic staining. It flows through a mixture of alluvial and colluvial geomorphology, with expansive gravel and cobble point bars more typical of relatively wide and shallow, higher gradient western North America rivers. An 8-m high falls, 4-km upstream from Lake Superior, is a barrier to upstream movement of fishes. Like other Lake Superior tributaries, the Cypress River is used by migratory salmonid species for spawning, rearing, and opportunistic feeding in the spring and autumn but becomes too warm and dry in July–August to sustain much more than juvenile salmonids. In addition to a spring spawning run of longnose and white suckers, a large number of rainbow smelt (*Osmerus mordax*) and approximately 1500 adult steelhead (*Oncorhynchus mykiss*) use the river to spawn in spring (Ontario Ministry of Natural Resources and Forestry, OMNRF, unpubl. data). Another estimated 1000 pink salmon (*Oncorhynchus gorbuscha*), with a much smaller number of brook trout (*Salvelinus fontinalis*), coho salmon (*Oncorhynchus kisutch*) and chinook salmon (*Oncorhynchus tshawytscha*) use the river for spawning in the autumn.

Sucker abundance

To estimate the abundance of spawning longnose and common white suckers, two 1.8 m high trap nets were placed near the mouth of the Cypress River in 2014. Traps were checked daily from 9 May to 3 June 2014. Suckers were marked with a fin hole punch specific to a time period (t) that changed roughly each week: upper caudal, dorsal, anal, right pectoral. Recaptured fish received an additional hole punch specific to the time period captured. Although this approach does not allow individual identification, it is fast and tags loss is not an issue. Capture data were analysed using program MARK V8.0 Jolly–Seber statistic for open populations (White & Burnham, 1999) following Schwarz *et al*. (1993). The prerequisites for the Jolly–Seber-statistic are

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as follows: (i) every individual has the same probability of being caught in the t sample, whether it is marked or unmarked, (ii) every marked individual has the same probability of surviving from the t to the $t + 1$ sample implying mortality and emigration as a combined term, (iii) individuals do not lose their marks and marks are not overlooked at capture and (iv) sampling time is negligible in relation to intervals between samples. No tests of equal catchability were applied. Survival was assumed to be very high because of the short duration of the experiment which was prior to spawning. The Mark loss was assumed to be 0%. On 27 May, with polarised glasses and dipnets, we walked two small neighbouring rivers, MacInnes Creek (N 48.931°, W 87.924°) and the Little Cypress River (N 48.925°, W 87.845°), from the TransCanada highway to Lake Superior to determine if suckers we tagged migrated to a different river. A combination of river surveys and angler reports were used to assess when suckers entered the Cypress River. Two 2-km surveys along the river and casual observations while working on the river were used to determine carcass counts and assess spawning mortality.

Resource subsidy: eggs

We estimated the mass of sucker eggs deposited in the river ($Eggs_x$), where N_x is the population estimate of species x, Sr is sex ratio, Mf is average mass of a female, and GSI is the gonadosomatic index as,

$$\mathrm{Eggs}_x = N_x \times Sr_x \times Mf_x \times \mathrm{GSI}_x \quad (1)$$

Mean fish mass, sex, and GSI were estimated from a subsample of each species taken just before the peak of the run. Because some fish may not shed all their eggs during spawning, we also calculated GSI at lower percentages of body mass. External characteristics such as nuptial tubercles were used to distinguish males from females. Length (±1 mm), sex (n = 503) and mass (±5 g, n = 278) were determined from a subsample of suckers caught in trap nets. A subsample of 78 fish (46 longnose and 32 white suckers) was used to determine GSI and fecundity. In the laboratory, ovaries were weighed (±0.1 g) and GSI estimated as (ovary mass/fish mass) × 100.

To estimate N and P contained in eggs, we used the findings of Childress & MacIntyre (2014). They determined that dried white sucker eggs were 10.7% N and 0.1% P and dry mass was 36.3% of wet mass. We estimated the amount of milt produced by longnose and white suckers using published GSI of 4.83% for male white suckers (Bowron, 2008). Total milt added to the river was estimated using eqn 1, but substituting the average male mass and GSI. Values for N (4.4%) and P (1.0%) contributions from milt were obtained from Gende *et al.* (2004). Carcass N and P were estimated at 10.4% N and 2.2% P of dry mass as determined by Childress & MacIntyre (2014). Dry mass was assumed to be 29.0% of wet weight (Cummins & Wuycheck, 1971).

Resource subsidy: excretory output and nutrients

Nutrient additions via sucker excretion, including SRP and ammonium (NH_4) were estimated in two ways: (i) sampling ambient water from the Cypress River, and (ii) directly estimating excretion from suckers.

Ambient water was sampled every 3 days from 10 May to 3 June 2014 spanning the pre- and post-spawn conditions. Water samples were collected 400 m upstream and 3700 m downstream (300 m upstream of Lake Superior) of the falls, representing control and impact respectively. Water samples were frozen for later analysis at the Lakehead University Centre for Analytical Services. Samples were thawed and filtered (Whatman GF/F, 0.7 μm). Ammonium (NH_4) was determined by chromatography, using a Dionex120 Ion chromatograph (Dionex Corporation, Sunnyvale, CA, U.S.A) and SRP by colorimetry via phosphomolybdic blue complex measured at 880 μm.

Nutrient subsidies were estimated experimentally in the field by conducting static tests where individual suckers (longnose n = 24, common white n = 14) were held in containers for 30–60 min. Mean water temperature during the tests was 12.8 °C and varied from 12.7 to 14.2 °C. Water was sampled from each 20 L container at the beginning and end of each test. The activity levels of suckers in the containers was likely lower (static) than that experienced in the river were they would be actively swimming against the current and interacting with other suckers; however, stress from capture and holding may have increased the metabolic output. All fish were measured (mass and total length) after the experiment. We used the linear regression to relate the total length to the excretory output (mg h^{-1}). A linear fit was more suitable than a power function due to the relatively small range of fish sizes. Excretion of SRP and ammonium was estimated for an average size sucker, and then multiplied by the population estimate for longnose and white suckers to give a daily total amount of NH_4 and SRP added to the river. Based on the presence of suckers in the river, we estimated that the spawning lasted 10 days (May 24 – 2 June 2014) and this was used

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to estimate total output of ammonium and SRP for the spawning run. The average water temperature during this period was 12.5 °C and thus very similar to water temperature during the static tests. For comparison, we also used the excretion equations of Childress & MacIntyre (2014) to estimate total NH_4 and SRP released into the river.

We also estimated what ambient river concentrations of NH_4 and SRP should be based on daily discharge and amounts of NH_4 and SRP added to the river by the spawning population for the same 10-day period. Stream nutrient concentrations below the expected values likely indicate nutrient uptake. Water level and temperature in the Cypress River was continuously monitored, using a Solinst levelogger (Solinst Canada, Georgetown, ON, Canada). An additional barologger provided atmospheric barometric correction. Water level estimates were converted into discharge estimates by using WinXSPRO V3.0 (Hardy, Panja & Mathias, 2005) which produces a stage–discharge relationship based on water surface slope, substrate size (d84), and channel cross section. Periodic discharge measurements provided calibration/validation.

Trophic transfer and incorporation of subsidy

Drift. All field sampling to examine the transfer and incorporation of subsidies in the Cypress River were conducted in 2013. Large numbers of suckers were observed spawning on 24 May. During the spawn, drift nets (1 mm mesh, 305 × 457 mm) were used to quantify the drift of eggs downstream in ten areas in the lower 2 km of river. Nets were set for 30–60 min in water averaging 0.4 m deep and a water velocity of 0.7 m s^{-1}. We calculated egg drift density (no. of. egg 100 m^{-3}), egg drift biomass (egg biomass 100 m^{-3}), and the number of eggs captured per hour in the net (no. of. egg h^{-1}), as an estimate of potential food availability for a resident juvenile trout or other eggs consumer (e.g. sculpin) in the Cypress River.

Basal resources, benthos, and fishes. In mid-July, 7 weeks after spawning, we sampled epilithon, benthic invertebrates and fishes below the falls at 10 sampling sites spaced at 500 m intervals. Above the falls, eight sites were located at 150 m intervals. Biomass of epilithon was estimated by scraping a 34 cm^2 area from the upper surface of two randomly selected cobbles from each site. Cobble scrapings were removed using a knife and then placed into a plastic vial and frozen. In the laboratory, macroinvertebrates were removed from the scrapings, which were then filtered on pre-weighed glass filters, dried at 40 °C, weighed, ashed at 550 °C for at least 3 h, then reweighed (±0.01 mg) to calculate ash free dry mass. Values were then converted into an aerial basis (mg cm^{-2}).

At the same sites, benthic invertebrates were sampled to a substrate depth of 10 cm using a 0.09 m^2 Surber sampler fitted with a 500 μm mesh and immediately preserved in 10% buffered formalin and later 70% ethanol. An effort was made to sample areas with similar depths, velocity, and substrate size composition to reduce variability from other potentially confounding factors. At each site, two benthic samples were collected in riffle areas. Water depth (cm), velocity (m s^{-1}; measured using a Swoffer 2100, Swoffer Instruments Inc. Seattle, WA), and substrate size composition were recorded at each sampling point. Substrates were classified as clay and silt (<0.0625 mm), sand (0.0625–2 mm), gravel (2–64 mm), cobble (64–256 mm), and boulder (>256 mm). In the laboratory, invertebrates were identified to genus and density estimates were calculated as individuals per square metre. Coarse particulate organic matter in the sample was retained and filtered through a 1-mm sieve to determine CPOM dry mass. CPOM was then dried for 12 h at 60 °C in a drying oven to a constant mass (±0.01 mg).

On the same day as benthic invertebrate surveys, fish communities were sampled using a Smith-Root LR24 backpack electrofisher (Vancouver, WA), using the single pass method. Downstream of the falls, five sampling sites were spaced at 1000 m intervals (every other benthic survey site). Above the falls, four sites were located at 300 m intervals (every other benthic survey site). Captured fish were identified, enumerated, weighed (±0.01 g) and measured (fork length ±1 mm). Area (m^2) electrofished was determined as the length of the site multiplied by the average width of three transects. Standing fish biomass was determined as fish biomass by area fished (wetted width × length). We used *t*-tests or Mann–Whitney Rank Sum Tests depending on normality to determine if standing biomass, benthic invertebrate density, CPOM and epilithon biomass were significantly different above and below the falls.

Stable isotopes. We sampled stream fishes and benthic invertebrates in summer and in autumn (49 and 117 days after spawning) to test for seasonal changes in isotopic ratios, $\delta^{13}C$ and $\delta^{15}N$. During these time periods, water temperature averaged 18.5 °C and the median mass of fish sampled was 1.2 g in July, which according to Thomas & Crowther (2015) would require

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24 days for 50% replacement of an isotope or 95–119 days for complete isotopic replacement. This estimate also concurs with laboratory experiments we conducted where brook and rainbow trout were fed sucker eggs in a hatchery system (N. E. Jones, unpubl. data). Samples were taken above and below the falls to evaluate the influence of migratory subsidies on isotopic ratios. Slimy sculpin and longnose dace were the most abundant species and were split into small (<60 mm) and large (>60 mm) groups. Sample sizes for the summer and autumn, both above and below the falls ranged from 14 to 33 individuals for each species. In total, 231 sculpin and longnose dace were analysed. We also sampled a small number of young-of-year brook trout and juvenile rainbow trout during the summer. Three adult adfluvial brook trout were sampled in the autumn to establish isotopic ratios for this fish species in Lake Superior. Benthic invertebrates, with a focus on Aeshnidae, Gomphidae, Heptageniidae, Ephemerellidae, Philopotamidae, Hydropsychidae, and Perlidae, were sampled in spring and summer above and below the falls. Sample sizes for invertebrates ranged from 20 to 28 individuals. Samples of longnose sucker eggs were also analysed. All stable isotope samples were frozen shortly after sampling in the field. Stable isotope analyses were conducted at the Water Quality Centre at Trent University, Canada. Isotope ratios are expressed as delta values (δ) and measured as the difference (in parts per thousand, ‰) between the isotope ratio of a sample and that of an international standard according to the formula: $\delta^{13}C$ or $\delta^{15}N = (Rsample - Rstandard/Rstandard) \times 1000$, where $R = {}^{13}C/{}^{12}C$ or ${}^{15}N/{}^{14}N$. Isotope ratios were determined, using an EuroEA3028-HT EuroVector Elemental Analyzer (EuroVector SpA, Milan) to burn the solid samples to CO_2 and N_2 gas followed by a Micromass IsoPrime Continuous Flow Isotope Ratio Mass Spectrometer (Cheadle Hulme, Stockport) to measure the isotope ratios of the gases. The reference material standard used for normalisation of the data was USGS40 ($\delta^{15}N = -4.56$ ‰; $\delta^{13}C = -26.39$ ‰). Casein (Elemental Microanalysis Ltd, Okehampton, UK) and DG(A1) (Sigma-Aldrich Canada, Oakville, ON, Canada) were working standards used for quality control. Repeat analyses of samples of fish and invertebrates demonstrated that analytical error was 0.2 and 0.3‰ for carbon and nitrogen respectively.

A four-way ANOVA was used to detect differences in fish stable isotope ratios among the fixed factors location (upstream and downstream), season (summer and autumn), size of fish (large and small), and species of fish (longnose dace and slimy sculpin). Brook trout were removed from this analysis because of the unbalanced design. For benthic invertebrates, a two-way ANOVA was used to detect differences in stable isotope ratios among the fixed factors location (upstream and downstream), and season (spring and summer). Significant main effects were further assessed with Tukey multiple comparison procedure. The Kolmogorov–Smirnov test was used to examine data for normality and the Levene median test for homogeneity of variances.

To estimate the proportions of prey items (sucker eggs and benthic invertebrates) in the diets of slimy sculpin and longnose dace, we used a Bayesian mixing model (MixSiar in R V3.0.2, Stock & Semmens, 2015). This model allows sources of uncertainty, as standard deviations around the food sources, to be propagated through the model to return a probability distribution of estimated dietary proportions. We used discrimination factors of 3.4‰ (SD = 0.2) for $\Delta\delta^{15}N$ and 0.8‰ (SD = 0.4) for $\Delta\ \delta^{13}C$ (Vander Zanden & Rasmussen, 2001). Invertebrate stable isotope values were pooled across spring and summer and across taxa ($\delta^{13}C$ −29.89 ‰, SD = 2.65 and $\delta^{15}N$ 3.37‰, SD = 0.99, $n = 47$). Isotope values for collected sucker eggs were ($\delta^{13}C$ −23.55‰, SD = 1.15 and $\delta^{15}N$ 7.17‰, SD = 0.23, $n = 10$). The model was run with a Markov Chain Monte Carlo settings of a chain length of 100 000 with a burn in of 50 000 and Resid × Process error selected.

Resource subsidy: other species. To assess the importance of sucker subsidies relative to those provided by other fishes, particularly non-native salmonids, we estimated contributions of carcasses, eggs, and milt throughout the year from all large migratory fishes including smelt. We used estimates of fish size, GSI, and abundance from published and unpublished data to estimate subsidies contributions. Given a paucity of data for fish abundance in the river, we relied on a mix of quantitative and qualitative information to provide best approximate values. Exact numbers were not required to provide a reasonable understanding of the relative importance of suckers. We assumed a 50 : 50 sex ratio, all eggs and milt were deposited in the river, and that carcasses remained in the river. We used an average mass for males and females.

Adfluvial spawning steelhead in the Cypress River represents a potentially large subsidy and their abundance is estimated annually via a mark recapture data Peterson estimate (J. George, OMNRF, pers. comm.). We used abundance estimates, sex ratio, and fork length data from the 2013 steelhead sampling campaign. GSI was estimated for male and female steelhead based on

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104 fish from the Steel and Goulais rivers, two tributaries of Lake Superior (T. A. Johnston, OMNRF, unpubl. data). We estimated the abundance of Pink salmon (n = 1000) and smelt (n = 30 000) that use the river based on observations and discussions with local agency biologists. Pink salmon GSI was estimated based on 48 fish from the St. Mary's River, Lake Huron. Smelt GSI estimates were taken from Lischka & Magnuson (2006). The abundance of other fishes including chinook and coho salmon, and brook trout is low. Smelt size was estimated in 2014 based on trapnet captures (current study) and pink salmon mean mass in the Cypress was estimated in 2013. Average fish mass and GSI estimates for chinook, coho, and brook trout were collected from Colvin, Close & Hassinger (1985), Stauffer (1976), Noaksson *et al.* (2003), and Newman, Dubois & Halpern (2003). GSI for female white suckers was determined in the current study and is very similar to that estimated by Bowron (2008) from Mountain Bay which is 3.5 km east of Cypress River. Bowron (2008) also provided GSI for males. Values for N (4.4%) and P (1%) contributions from milt were obtained from Gende *et al.* (2004) for all fishes. We also used Gende *et al.* (2004) nutrient values for carcasses and eggs for all fishes (carcass N 2.41% P 0.41%, eggs N 4.4% P 0.35%) except suckers.

Results

Only thirty-seven suckers were caught during the first 7 days, until 16 May when 624 suckers were captured. Lake water temperature was 2.5 °C while maximum daily water temperature in the river was 3.2 °C. Catches increased markedly with post-freshet decreases in flow and increasing water temperatures (Fig. 1). Catches peaked on 23 May with 4123 new suckers when the daily maximum water temp was 10.3 °C in the river and 3.9 °C in the lake. Based on our observations, and that of steelhead anglers staying at our field camp, suckers accumulated at the river mouth until 25 May. In total 22 290 suckers, 16 057 longnose and 6233 white, were captured and marked over the course of the trap netting period. A total of 2778 suckers were recaptured, 2123 longnose and 655 white. For the population estimates, we used data collected up to and including 28 May because after this date we noticed increases in the numbers of spent fish and catch rates began to increase again signalling the out migration of fish (Fig. 1). The population estimate for both species of sucker was 82 449 (95% CI = 75 885–89 733), 67 811 longnose sucker (95% CI = 63 565–72 340) and 14 639 white sucker (95% CI = 12 320–17 394). Using the length of the river (4 km) and wetted width (10 m at a discharge of 10 $m^3 s^{-1}$), density was estimated to be approximately two suckers per square metre of riverbed.

Resource subsidy: carcasses, eggs, and milt

Mean female longnose sucker fork length and mass were 444 mm (SD = 35) and 1061 g (SD = 237), and based on fecundity estimates contained 16 558 eggs, with an average GSI of 12.7% (SD = 3.4). Mean female white sucker fork length and mass were 448 mm (SD = 33) and 1007 g (SD = 217), and based on fecundity estimates contained 14 301 eggs, with an average GSI of 12.0%

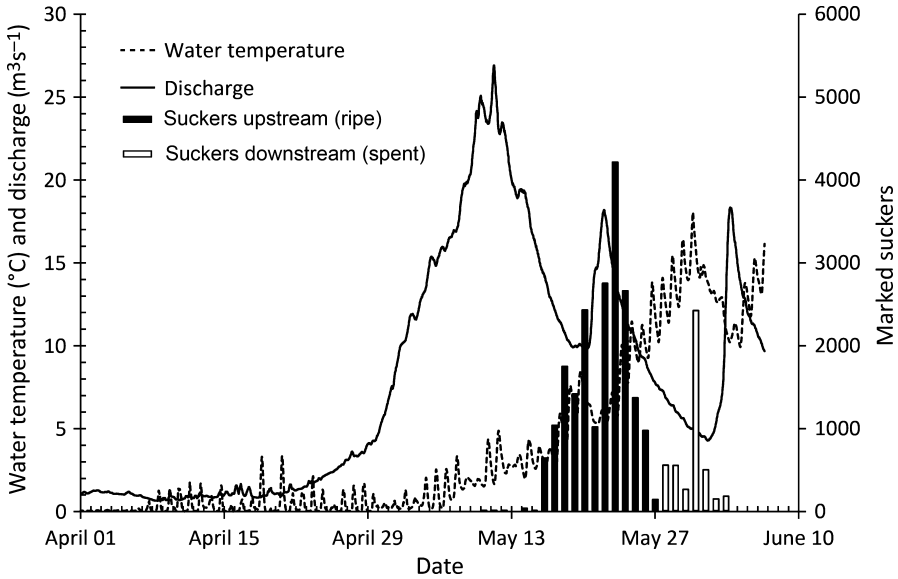

Fig. 1 Discharge and water temperature in relation to the number of longnose and white sucker captured in trapnets near the mouth of the Cypress River. Pre-spawn suckers moving upstream from 16 May–28 May and spent sucker moving downstream 29 May–3 June. Daily trap netting began on 9 May.

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(SD = 2.6). Eggs for both species averaged 3.1 mm (SD = 0.31) in diameter and had a mass of 8.5 mg (SD = 1.72). We calculated that 5635 kg of eggs were deposited at a GSI of 13%. Because some fish may not shed all their eggs during spawning, GSI values of 12, 11, and 10% values provide egg mass estimates of 5201, 4768, and 4334 kg respectively. Using the lower and upper 95% confidence levels of abundance (i.e. 75 885–89 733 fish) and a GSI of 13%, the estimated mass of eggs deposited ranged from 5190 to 6127 kg. The addition of milt to the river was substantial with longnose sucker males adding an estimated 1650 kg, 95% CI = 1629–1759 and white sucker males adding 375 kg, 95% CI = 316–423. One common white sucker, showing no signs of injury, was found dead in the river during carcass surveys.

In total there was an estimated 5934 kg of eggs, 2123 kg of milt and 1275 kg of carcasses from the eight species of migratory fish that use the Cypress River (Table S1). Relative to other mainly non-native fishes, suckers provided 92% of this annual egg biomass and 95% of the milt. Suckers, however, provided <1% (1 kg) of the annual carcass biomass, whereas, non-native pink salmon provided 50% (600 kg). Most of the annual subsidy is delivered during April, May and June when the suckers spawn. Based on these estimates, suckers provide 84% and 78% of the annual subsidies of N and P, or 303 and 28 kg respectively.

Resource subsidy: excretory output and nutrients

Analyses of ambient river water chemistry indicated that concentrations of NH_4 and SRP for many samples (21 out of 32) were below minimum detection levels 0.01 and 0.0050 mg L^{-1} respectively, both upstream and downstream of the falls, both before and after spawning (Fig. 2).

The concentration of NH_4 was below detection limits for approximately half of the samples used to establish pretest water chemistry for the static tests. In such cases, we assumed a minimum concentration of 0.01 mg L^{-1} for NH_4. Static tests indicated that an average-sized (450 mm total length) sucker produced 8.88 mg h^{-1} of NH_4 based on the regression relationship NH_4 mg h^{-1} = 0.0429 × (total length mm) − 10.422, $R^2 = 0.28$, $P < 0.001$. Similarly, an average-sized sucker produced 0.57 mg h^{-1} of SRP based on the regression relationship SRP mg h^{-1} = 0.0017 × (total length mm) − 0.194, $R^2 = 0.38$, $P < 0.001$. At these rates of excretion and given the spawning population size, we estimated that

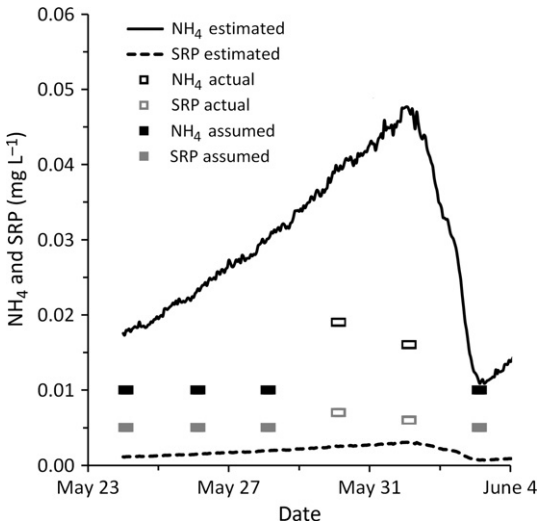

Fig. 2 Nutrient concentrations of ammonia and soluble reactive phosphorus as measured in ambient stream water and estimated based on excretion rates of individual suckers and discharge. Concentrations in some samples were at or below detection levels. All samples from above the falls were at or below detection levels.

212 kg NH_4 (95% CI: 195–230) and 14 kg SRP (95% CI: 13–15) were released into the Cypress River over the 10-day spawning period. Using the excretion equation of Childress & MacIntyre (2014) an estimated 180 kg NH_4 (95% CI: 165–195) and 19 kg SRP (95% CI: 18–21) was released into the Cypress River.

Ambient NH_4 concentrations in the river, both assumed and measured minimums, were consistently below that predicted using excretion values for an average-sized sucker, the population estimates, and discharge over a 10-day period (24 May – 2 June; Fig. 2). Conversely, the ambient SRP concentrations were consistently above that predicted although the general patterns were similar over the spawning period. The higher ambient SRP values are likely due to our assumption that samples below detection were at the detection limit (0.05 mg L^{-1}) when they may have been much lower and closer to that predicted.

Trophic transfer and incorporation of subsidy

The number of eggs drifting downstream into the net was 479 h^{-1} (Fig. 3). Egg density was 136 eggs 100 m^{-3} and biomass of eggs was 2.7 g 100 m^{-3} (Fig. 3). Epilithon mass was over nine times higher downstream (2.27 mg cm^{-2}, SE = 1.09) than upstream (0.25 mg cm^{-2}, SE = 0.09) of the falls ($U = 36$, d.f. = 16, $P = 0.006$, Fig. 4). The dried biomass of CPOM was no different downstream (5.4 g m^{-2}, SE = 1.44) than upstream (6.5 g cm^{-2},

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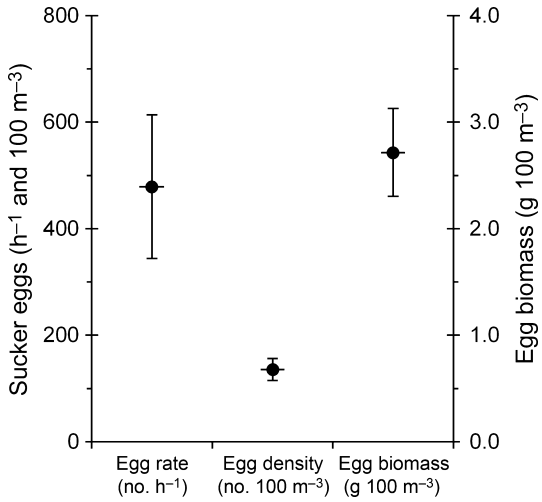

Fig. 3 Density and biomass of sucker eggs drifting downstream in the Cypress River. Errors bars are standard errors.

SE = 1.56) of the falls (t = −0.797, d.f. = 16, P = 0.437). Benthic invertebrate densities were more than two times higher downstream (2899 ind. m^{-2}, SE = 664) than upstream (1249 ind. m^{-2}, SE = 207) of the falls although densities did not differ significantly due to a high degree of variability and small sample size (t = 1.94, d.f. = 16, P = 0.070, Fig. 4). Fish standing biomass was eight times greater downstream (264 g m^{-2}, SE = 89.9), than upstream (32 g m^{-2}, SE = 13.9) of the falls (t = 4.16, d.f. = 6, P = 0.005, Fig. 4). Among sites, longnose dace and slimy sculpin represented 66% of the fish biomass.

Consistent with a resource subsidy, fishes downstream had significantly higher δ^{13}C values than fish upstream (DS −23.2‰, SD = 1.73, US −25.8‰, SD = 1.08; P < 0.001; Fig. 5). Also of interest, there were no consistent differences in δ^{13}C values among seasons (SU −24.4‰ SD = 1.99, FA −24.5‰ SD = 1.44; P = 0.87) or between slimy sculpin and longnose dace (LND −24.2‰ SD = 1.60, SC −24.8‰ SD = 1.72; P = 0.161). Large fishes (>60 mm) had higher δ^{13}C values than small (<60 mm) fishes (L −24.4‰ SD = 1.70, S −25.2‰ SD = 1.35; P = 0.98). Sucker eggs were had δ^{13}C values than sculpin and dace downstream of the falls in the summer (eggs −23.6‰ SD = 1.15, fish −22.9‰ SD = 1.26, P = 0.025).

Also consistent with a resource subsidy, fishes downstream of the falls had significantly higher δ^{15}N values than fish upstream (DS 7.5‰ SD = 0.88, US 7.0‰ SD = 0.64; P < 0.001; Fig. 5). Fishes in summer had higher δ^{15}N values than in the autumn season (SU 7.5‰ SD = 0.81, FA 7.1‰ SD = 0.97; P = 0.04). Large fishes had higher δ^{15}N values than small fish (L 7.6‰ SD = 0.97, S 6.9‰ SD = 1.02; P = 0.001). Slimy sculpin had significantly higher δ^{15}N values than longnose dace (SC 7.7‰ SD = 0.92, LND 6.9‰ SD = 0.71; P < 0.001). Sucker eggs had lower δ^{15}N values than sculpin and dace downstream of the falls in the summer (eggs 7.2‰ SD = 0.23, fish 7.9‰ SD = 0.75, P < 0.001). Large adflu-

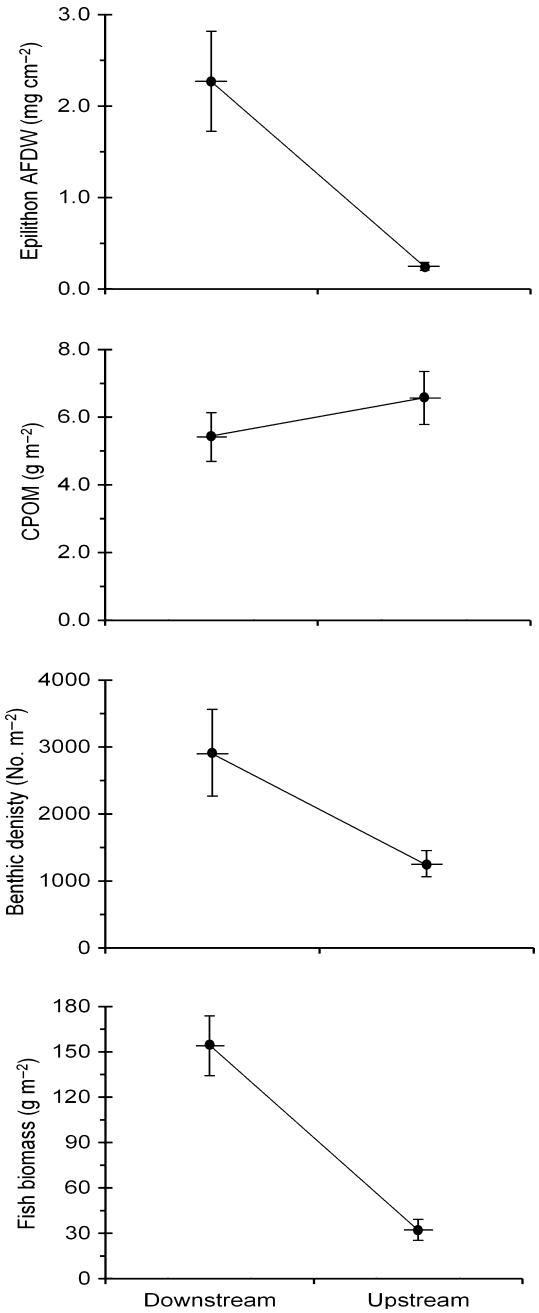

Fig. 4 Quantities of epilithon, coarse particulate organic matter, benthic invertebrates, and fishes upstream and downstream of the falls on the Cypress River. Downstream of the falls, suckers and other migratory fishes have full access and can provide resource subsidies. Errors bars are standard errors.

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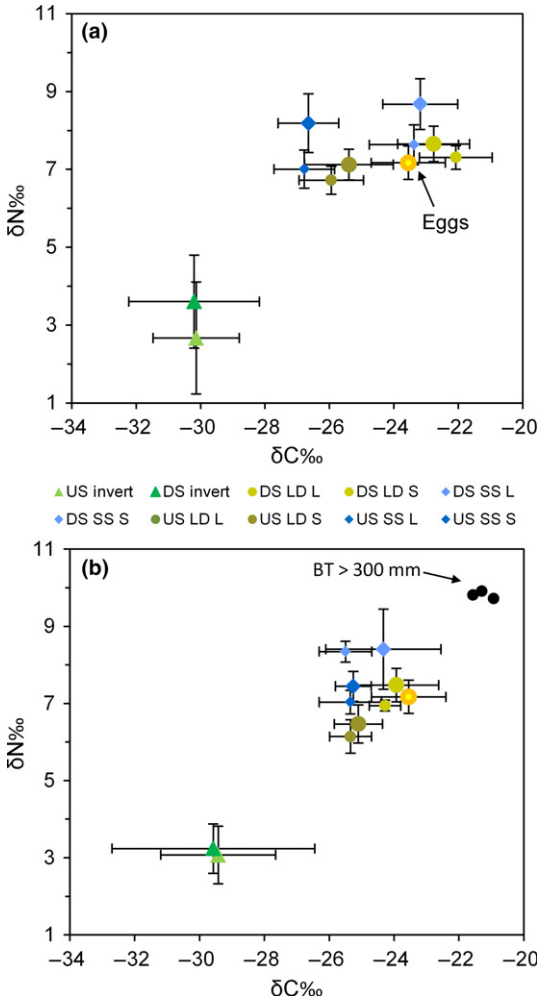

Fig. 5 Stable isotopes ratios of fishes in the summer (a) and autumn (b) sampled upstream (US) and downstream (DS) the falls on the Cypress River. Fish species include longnose dace (LD), slimy sculpin (SS). Isotopic values for sucker eggs are indicated by the large orange symbol and associated standard deviation error bars. Three large migratory brook trout >300 mm fork length (black circles) were also sampled to established an understanding of isotopic ratios from Lake Superior.

vial brook trout caught in the fall downstream of the falls had higher $\delta^{13}C$ values (−21.3‰, SD = 0.59) and $\delta^{15}N$ values (9.5‰, SD = 0.32) than all other fishes.

Benthic invertebrates downstream and upstream of the falls had similar $\delta^{13}C$ values (DS −29.5‰ (SD = 2.65), US −29.2‰ (1.61); $P = 0.506$). In contrast, benthic invertebrates downstream of the falls had significantly higher $\delta^{15}N$ values (DS 3.7‰ SD = 0.99, US 3.1‰ SD = 1.16; $P = 0.01$; Fig. 5) and no differences among seasons (SU 3.4‰ SD = 1.40), SP 3.3‰ SD = 0.70; $P = 0.400$). Invertebrates in the summer had higher $\delta^{13}C$ values than in the spring (SU −28.7‰ SD = 2.54, SP −30.1‰ SD = 1.72; $P = 0.007$); however, there was a significant interaction effect between location and season

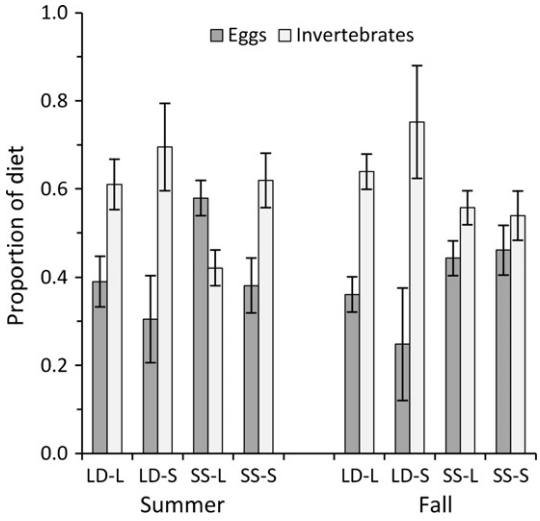

Fig. 6 Dietary proportions of sucker eggs and invertebrates of large (L) and small (S) longnose dace (LD) and slimy sculpin (SS) in the summer and autumn based on Bayesian mixing model (MixSiar). Errors bars are standard errors.

($P = 0.031$). Invertebrates downstream had higher $\delta^{13}C$ values in summer (SU −28.5‰ SD = 3.13, SP −30.7‰ SD = 2.03; $P < 0.001$).

Based on the MixSiar Bayesian mixing model, eggs were an important subsidy to the diet of both sculpin and longnose dace, particularly from fish sampled in summer (30–58%), less so in the fall (25–45%; Fig. 6). Eggs made up a greater proportion of the diet in sculpin than longnose dace and in large-sized fishes relative to small-sized fishes of both species.

Discussion

This study supports the hypotheses that adfluvial suckers provide a large material subsidy that increases the productivity of the Cypress River and that this resource subsidy exceeds that delivered by other native and non-native fishes (e.g. Pacific salmonids). Our results generally agree with recent stream nutrient-enrichment research conducted in the Great Lakes Basin in which spawning suckers added nutrients and eggs that elevated productivity at multiple trophic levels (Childress & MacIntyre, 2014; Childress *et al.*, 2014).

The Cypress River received approximately 212 kg NH_4 and 14 kg SRP during the spawning period not including nutrient output directly from eggs. Ambient NH_4 concentrations were consistently well below that predicted from excretion equations suggesting that much of the released ammonium is rapidly taken up by river microorganisms. Similar uptake by algae and biofilm

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has been well noted in salmon subsidy systems (e.g. Wipfli, Hudson & Caouette, 1998; Claeson *et al.*, 2006) and sucker systems (Childress *et al.*, 2014) and relates well to the much higher amounts (9×) of periphyton found in the subsidised portion of the Cypress River. Ambient SRP concentrations in the Cypress River were consistently above that predicted although the general patterns were similar over the spawning period. The higher ambient SRP values may be due to our assumption that samples below detection were at the detection limit when they likely were much lower and typical of oligotrophic streams. Nonetheless, phosphorus contributions from suckers can provide a significant portion of the annual budget, particularly in phosphorus limited watersheds with elevated agricultural inputs of N (Childress & MacIntyre, 2014). In contrast, the productivity of nutrient poor Boreal rivers is often constrained by N availability (Burrows *et al.*, 2015). These differences suggest that the environmental context of nutrient limitation and the inputs from fishes will together dictate the relative importance of different nutrients (Childress & MacIntyre, 2014). In very productive rivers, light rather than nutrients may limit primary productivity (Rand *et al.*, 1992). Eggs, however, if directly consumed can increase secondary production even in nutrient-rich productive streams by injecting nutrients and energy at higher levels of the trophic pathway.

Suckers in the Cypress deposited between 5190 and 6127 kg of eggs resulting in a large number of eggs drifting downstream into drift nets (479 h^{-1}) which is likely far greater than the feeding and digestive capacity of a juvenile fish. Although we quantified the potential for a large subsidy of milt by suckers (2124 kg) we know little about actual amounts expelled and its fate in the river. Further investigation is warranted to understand the magnitude of the milt subsidy and how it is incorporated into the river food web. Whereas salmon carcasses represent a large and important source of nutrients and energy rich organic matter in streams (Wipfli *et al.*, 1998; Gresh *et al.*, 2000; Gende *et al.*, 2002) we found a small number of sucker carcasses in the Cypress River, indicating that they contribute little to the material subsidy.

There were no differences in CPOM biomass upstream and downstream of the waterfall which suggests that increased invertebrate production is the result of other sources of energy such as biofilm, periphyton, fish eggs and carcasses. Higher benthic densities in the subsidised portion of the Cypress River parallels the findings of many others studies in salmon subsidised systems (e.g. Kline *et al.*, 1990; Wipfli *et al.*, 1998) and sucker subsidised tributaries (Childress & MacIntyre, 2014).

Higher fish biomass downstream of the falls echoes the observations of others examining the influence of salmon subsidies on juvenile fish density, lipid content, biomass, and growth (Michael, 1995; Bilby *et al.*, 1998). Eggs can provide resident stream fishes with much of their annual energy surplus even if the subsidy comes as a pulse during 5 weeks (Armstrong & Bond, 2013). The digestive flexibility of Dolly Varden (*Salvelinus malma*) suggests an adaptability to rely on an annual pulse of eggs that is predictable (Armstrong & Schindler, 2011; Armstrong & Bond, 2013). The importance of an annual subsidy may have stronger and unanticipated impacts on fish growth and overwintering survival (Ward & Slaney, 1988; Shuter & Post, 1990; Biro *et al.*, 2004).

Isotopic ratios provide a time integrated representation of diet reflecting the movements of fish or the transport and incorporation of materials across ecosystem boundaries (Rasmussen, Trudeau & Morinville, 2009). The disequilibrium in stable isotopes we observed between upstream/downstream benthic fishes is consistent with a material subsidy as shown by others (e.g. Kline *et al.*, 1990; Bilby *et al.*, 1996). Biota upstream of the falls generally had the lowest $\delta^{13}C$ and $\delta^{15}N$ values, whereas, large migratory brook trout caught downstream of the falls had higher $\delta^{13}C$ and $\delta^{15}N$ values than all other fishes reflecting their lake-derived signature. Resident fishes downstream of the falls showed intermediate $\delta^{13}C$ and $\delta^{15}N$ values reflecting a mixed diet (i.e. sucker subsidy from Lake Superior and in-stream forage). Invertebrates did not show similar isotopic shifts as fishes. This may be because of the diversity of invertebrates, and thus functional feeding groups, analysed in our study. A smaller set of invertebrates could have been used with some prior knowledge of fish prey items. Isotopically rich sucker eggs, which were estimated to comprise 25–58% of the diet of fishes, increased isotopic values in fishes. In an examination of habitat use and trophic ecology of brook trout in Lake Superior tributary streams, Robillard, McLaughlin & Mackereth (2011; Fig. 4) found that fish above impassable barriers (residents) had lower isotopic values, whereas, fish below barriers had higher isotopic values consistent with feeding in the lake or a subsidy from adfluvial fishes. Perhaps values of $\delta^{13}C$ and $\delta^{15}N$ in stream fishes might be a useful index of the significance of lake derived subsidies not only in the Cypress River but also other tributaries in the Laurentian Great Lakes.

The importance of sucker subsidies may rival that of salmon related to key differences in the type, magnitude, and timing of the subsidy. Unlike salmon, few suckers

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die after spawning to leave carcasses for consumption and decay, however, the number of suckers spawning generates a significant mass of material. Eggs are high in energy (Cummins & Wuycheck, 1971) and contain numerous essential fatty acids (Schindler *et al.*, 2003). Unlike a salmon carcass, sucker eggs are immediately available to fishes and considerably smaller than the eggs of most salmonines (e.g. 3 versus 6 mm) making them available to smaller fishes that may otherwise be gap limited (Keeley & Grant, 2001). Sucker eggs are broadcast over riffles without nest building or parental guarding which contrasts with most salmonines where eggs are protected under substrate and sometimes guarded by females (Greeley, 1932; *cf* Bentley *et al.*, 2012). The absence of redd building also means less bioturbation of substrate and disturbance of invertebrates (Moore & Schindler, 2008; Albers & Petticrew, 2012). In addition to differences in subsidies between salmon and suckers, they also spawn at different times of the year. Salmon carcasses arrive in the fall when light and water temperatures are quickly decreasing, and thus, the subsidy might not be incorporated effectively. In contrast, suckers spawn when light and water temperature are increasing daily, deciduous riparian vegetation has yet to leaf-out meaning more sun for photosynthesis resulting in rapid uptake of nutrients and biofilm production. Lastly, upon hatching, sucker fry soon leave the river and thus remove little nutrients from the stream. Conversely, salmon and trout may spend 1–3 years as juveniles feeding in the parent river, and thus, export some nutrients, particularly if they die in the lake/ocean and never return.

Dams, barriers, and obstacles can prevent upstream fish migration leading directly to decline or loss of fish species that depend on longitudinal movements along a river during various life history stages (e.g. spawning and rearing habitats; Fagan, 2002; Harford & McLaughlin, 2007; Fausch *et al.*, 2009). Whereas some barriers prevent all species from upstream migration, and thus all potential subsidies, some barriers are selective, passing some fish species but blocking others depending on a variety of factors such as fish species, fish size, flow magnitude and water temperature. For example, low-head barrier dams common around the Laurentian Great Lakes, are designed to prevent passage of non-native sea lamprey (*Petromyzon marinus*) while allowing the upstream movement of most salmon, char, and trout. Unfortunately, non-jumping species including suckers are stopped as are the nutrients and eggs they would have delivered upstream. The loss of the material subsidy has been called oligotrophication (Stockner & Ashley, 2003; Anders & Ashley, 2007) and may lead to decreased production of juvenile fishes perhaps limiting their abundance, growth and overwinter survival (Biro *et al.*, 2004), particularly in already low productivity systems. In river systems that have lost stocks of salmon, production and survival of juvenile salmon can be low leading to negative feedback loop and the stream becomes 'stuck' in a low productivity state (Compton *et al.*, 2006).

The migration of iteroparous potamodromous adfluvial fishes can provide large and important nutrient subsidies to stream ecosystems, particularly when the key conditions of large migrant biomass, high local mortality or nutrient release, and retention of subsidy in streams of low nutrient status are met. While the subsidy increased the productivity of the Cypress River, we do not know if phenologies of recipient biota are keyed to take full advantage of this donor-controlled resource or if food web stability is influenced by the subsidy (Knight *et al.*, 2005; Moore *et al.*, 2008; Richardson, Zhang & Marczak, 2010). Further work is needed to examine if there is some dependency on the subsidy contributing to the evolution of phenology or is the subsidy too unreliable or variable? This is particularly relevant for native fishes that have co-existed for millennia, but adds an interesting complexity when non-native salmon (e.g. pink salmon) also provide a subsidy. Further research should also examine how variability of spawning sucker abundance relates to river productivity, and at what abundance level does a river become saturated (Bilby *et al.*, 2001; Moore *et al.*, 2008). We contend that material subsidies (eggs, milt, and carcasses) of suckers and many other potamodromous fishes are likely significant and more widespread than currently understood.

Acknowledgments

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Table S1. Quantities of egg, milt, carcass, and total N and P subsidies from fishes using the Cypress River. Numbers in parenthesis are percentages.

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